

A Smart Pill Container For Accurate Pill Retrieval Detection And Disease Control

Runcong Wang, Qinghua Zhang, Xiaolin Li, Nan Zhou, Hanqi Chen, Zhengqi Zhang

✉ runcong.wang@tum.de, qinghua.zhang@tum.de

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Abstract — This report provides an improved solution to address the long-term tracking of a patient's medication status. By referring to existing design solutions, a smart pill Container that can record the medication taking status is designed. An Arduino and corresponding sensor components were utilized. And the measured data were analyzed and summarized at the end. These data provide strong support for the analysis of the incidence of a number of long-term diseases and for physicians' understanding of patients' medication regimens.

keywords: Pill Container; Long-term diseases ; Arduino ; Medication ; Adherence records

In this report, the second section will describe the motivation of the project, while the third introduce the existing smart pill container in the market and the development trend. The forth section will describe in detail the specific operation process of this experiment, its core methodology and experimental analysis results. The fifth section will give a comprehensive summary of this report and in the last section, it will propose the direction of improving the smart pill container in the future, so that its functions can be more precise and detailed.

2 Motivation

1 Introduction

Medication non-adherence is a significant problem in healthcare worldwide, which can lead to the gradual worsening of patient health and higher costs [1]. One study showed that nearly 50 percent of patients suffer from increased morbidity and mortality due to unregulated medication use, resulting in an annual economic loss of about \$100 billion [2]. For patients with cancer and cardiovascular disease, the economic loss to the entire healthcare system can be more than double if medication is not taken as prescribed [3]. To address this issue, various smart medication adherence monitoring devices have been developed to regulate the behaviour of patients in taking medication at regular intervals. Compared to traditional medication methods, smart pill bottles can effectively help patients take their medication on time and correctly. With various sensors on smart pill bottles, the needed information or changing in the pill bottles can be precisely detected and collected [4, 5, 6].

Internet of Things (IoT) [7] supports the transmission of patient medication records. The monitoring of medication taking can be online completed by healthcare [8]. This is very helpful in solving the problem of wrong medications caused by the elderly because they cannot contact the doctor in time.

Smart pill Container still needs further research and improvement in practical applications. It is well known that most patients suffering from chronic diseases, such as Alzheimer's, diabetes, heart disease, mental illness and other diseases, tend to take many different medications at once. However, many patients often forget to take their medications due to poor medication compliance and poor memory, or take the wrong number of medications at the same time, all of which can lead to very serious drug abuse problems. For elderly people who live alone, without reminders from family members, the situation of missing and taking the wrong medication is even more serious.

In practice, how to distinguish the behaviour of Alzheimer's disease patients who pick up the pill bottle without actually taking the medicine, and how to ensure that the smart pill bottle can accurately determine whether the patient is actually taking the medicine, are all situations that pose challenges for the smart pill bottle. In this report, the smart pill container developed by our group possesses two major innovations. Firstly, it improves the functionality of pill retrieval detection. The pill container is equipped with three kinds of sensors: built-in inertial sensors, light sensors and pressure sensors, which can monitor the real-time attitude change of the pill container, monitor whether the cap is opened or not, and judge whether the patient is holding the pill con-

tainer and completing the swallowing action or not, respectively. At the same time, the data collected by the sensors are further analysed and processed to benefit the control of the disease. On the other hand, the smart container we have developed, while improving the functionality, also reduces the cost of production, costing only €15 for the intelligent management of the bottle, which makes it more suitable for wider dissemination than other container on the market.

3 Related Works

Thangam et al. [4] introduced a smart pill box system, which is equipped with Raspberry Pi, a weight sensor and a camera. Besides the measurement of weight changes, computer vision techniques are leveraged to capture pill photos. Then the author used YOLOv5 Model to precisely recognize and categorize various pill types. Furthermore, this system enables to alert healthcare providers in case of discrepancies or missed doses. In this way, it can help doctors and patients to identify the pills and verify dosages. Medication non-adherence can be reduced.

Murtadha et al. [5] developed a low-cost and wireless smart pill-bottle system. The system uses inertial data to build a patient discriminator model with classification techniques. The approach is to extract hand movements of users when taking pills from a bottle and use body gestures to identify who took the pills. By integrating acceleration sensors in medication bottles, it frees the user from wearing any devices while maintaining high accuracy discriminating users.

Nasir et al. [8] developed an IoT-based medication monitoring system for elderly patient. Equipped magnetic switch sensor will detect whether patient open or close the pill-box cap. Additionally, the Force Sensitive Resistor sensor were applied to detect the differences in force or pressure of the pills during consumption. With the utilization of IoT (Internet of Things), caregiver can monitor their patients' medication intake without having to be with them all the time.

Zijp et al. [6] introduced a Smart Pill Bottle Prototype (SPBP), which integrates real-time monitoring and automated dispensing mechanisms to enhance medication adherence. Recording the exact time and quantity of medication dispensed can be implemented with the help of two LMT85-Q1 tem-

perature sensors: one internal and one external. In a two-week user study, the SPBP achieved an 88% adherence rate, with a System Usability Scale (SUS) score of 79.3. Compared to traditional adherence methods, the SPBP provides granular, real-time tracking and objective data, reducing reliance on self-reports.

Rania. A. Molla [9] has proposed a smart medication monitoring system that combines smart prescription containers with a mobile application called SPC App. The system allows the patient to enter multiple prescription information in the SPC App where each medication corresponds to a specific container with a unique ID. By presetting the medication information and dosing frequency in the mobile app SPC App, the bottle indicator lights up when it's time for the patient to take their medication to remind them to take it on time. In addition, the Weighting Scale at the bottom of the medicine bottle can detect the real-time drug taking situation to ensure that the patient has taken the prescribed amount of medicine, thus achieving the purpose of monitoring the patient's medication time, dose, and informing the patient of the refill time and expiration date.

DeMeo and Morena [10] proposed a mechanism to monitor the medication taken by patients to ensure that they take their medication as prescribed. They designed a smart pill bottle that monitors medication taking by sending data to a server for analysis through wireless communication technology. The pill bottle has three notification methods: phone call, SMS, and an indicator light and tone on the bottle. The system also has a call-back function. Once a patient is found to be chronically not taking their medication on time, the system automatically collects the patient's feedback and the reason for the missed dose via SMS or phone call, and transmits this information to the healthcare provider for the next step of human intervention. The smart pill bottle is available on demand and automatically connected to the cloud, which greatly improves patient convenience.

Rumi et al. [11] proposed a smart pill box applying Internet of Things (IoT) technology for intelligent medication management by scanning the key drug information of the prescription through the camera on the pill box and uploading it to the database. The pill box has an inbuilt buzzer to notify the patient to take the medicine in time, its external LCD screen displays the drug information, and the medicine can be taken only after fingerprint verification of the pa-

tient's identity. The hardware structure of the smart pill box is complex, which includes six separate dispensing tubes, a spring-loaded control device to prevent medication from falling, a fingerprint recognition sensor and a buzzer, among other accessories. The pill box will provide the exact dosage of prescribed medication based on cloud information and push the medication out of the dispensing holes through a spring mechanism to achieve intelligent and accurate dispensing. When there is insufficient medication in the dispensing tube, the system will send an alert message to the patient through the display screen, achieving semi-intelligent management.

Othman and Ek [12] presented another improved smart pill dispenser. The innovation of this study is that instead of being limited to the built-in alerts found in traditional pill bottles, it sends high-volume alerts to patients and family members via the Instapush app on smartphones, thus ameliorating the lack of reminders to take medication in older adults who may be suffering from hearing loss. The device combines physical and smart technology. When the user presses the dispenser button, a built-in vibrating motor shakes the pill to drop it in the tray. At the same time, an infrared sensor counts the pills to ensure that the patient takes the medication in accordance with the preset dosage. This design not only improves medication adherence among elderly patients, but also reduces health risks caused by incorrect dosage or insufficient reminders.

Peddisetti et al. [13] developed an IoT-based smart medication management system in order to improve patients' medication adherence. The system consists of a smart mug and pill dispenser connected to an Android application for convenient medication management. The technology can accurately provide patients with precise measurements of the medication they take and the time they take it, ensuring that patients take their medication as prescribed and enhancing the safety of healthcare.

Al Shammari et al. [14] presented the limitations of smart pill bottles and directions for improvement. Some pill bottles are designed to be allowed to hold only one medication and are not suitable for patients with multiple prescriptions. Others are opened out of curiosity and not taken, but are defaulted to by the smart bottle as having been taken, leading to incorrect assessments. For patients who do not want to disclose their health status (e.g., HIV positive), the smart pill bottle can reveal their privacy to a certain

extent, thus limiting the group of smart pill bottles that can be used.

4 Technical Details

4.1 Project Setup

In order to confirm whether the Pill Container is being picked up by the patient and whether the patient is taking the medicine, three different sensor elements are used in order to minimize the cost and increase the accuracy of data recording. They are the BH1750 Light Intensity Sensor which is responsible for detecting whether the cap of the Pill Container has been opened, the Flexible Thin Film Pressure Sensor which detects whether the patient is holding the Pill Container, and the ICM-20948 9-DoF IMU which measures the motion status and attitude of the Pill Container. Figure 1 illustrates a picture of the final smart pill bottle model.

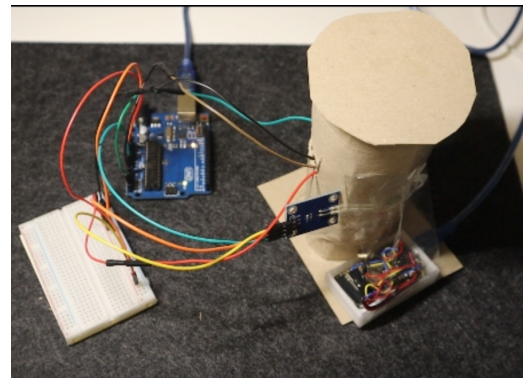


Figure 1 Smart Pill Container Test Model

4.1.1 BH1750 Light Sensor

The light sensor (BH1750) detects if the cap is opened. A highly sensitive BH1750 light sensor is placed inside the container. Once the cap is removed, the sensor registers an increase in light intensity, thus confirming that the bottle has been opened. In this way, we can determine when the patient has attempted to take the medication.

The BH1750 has a high sensitivity and can still accurately measure changes in light intensity in lux even in darker environments. Figure 2 shows the circuit of the BH1750 with the arduino.

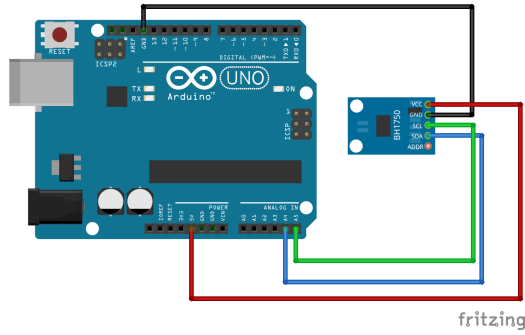


Figure 2 BH1750 Circuit

4.1.2 Flexible Thin Film Pressure Sensor

A pressure sensor is used to detect whether the pill container is being held.

The flexible film pressure sensor is mounted on the body of the container. It works by sensing the change in pressure on the side of the container when the patient holds it. By holding the container, the patient exerts pressure on the side of the container, which creates friction to overcome gravity to pick up the container. If no pressure is detected, the patient may not be holding the container at all. This step helps us eliminate accidental opening of the container cap and ensures that the patient is actively handling the container. We have also implemented a swallow confirmation mechanism.

After taking a pill, the patient must actively confirm that the pill has been swallowed by pressing the designated pressure sensor probe on the bottle. This additional step prevents false alarms, such as when a patient picks up the bottle but forgets to take the medication

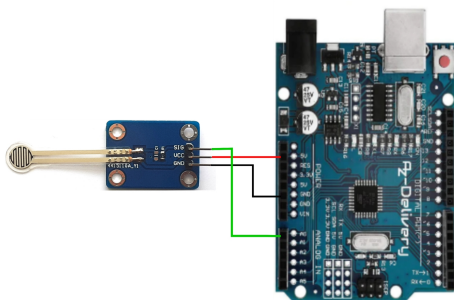


Figure 3 Flexible Thin Film Pressure Sensor

4.1.3 ICM-20948 9-axis Motion Tracking device

The IMU is then used to track the movement of the pill container.

When the patient picks up the pill container and prepares to take the medicine, the container moves through space. To capture this process, the ICM-20948 9-DoF IMU is employed. The ICM-20948 9-DoF IMU is based on the MPU9250 sensor, which includes a gyroscope and magnetometer. The sensor records the container's movement, tilt angle and acceleration, and is able to distinguish between a patient's casual maneuvering and an actual attempt to take the medication.

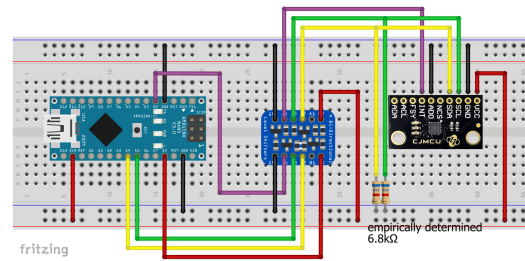


Figure 4 ICM-20948 9-axis Motion Tracking device

4.1.4 MPU9250 Sensor Fusion

In order to fuse the MPU9250 gyroscope and magnetometer data, a Madgwick and Mahony filter was utilized for data calibration and fusion with reference to the method of Kris Winer et al.[15] The fusion process is given in Fig5

Euler Gyro: Module that receives angular velocity and calculates the euler angle.

Euler Accel: Module that receives angular velocity and calculates the euler angle.

Madgwick and Mahony filter algorithms: Module that calculates the margin angle by combining data from the gyro sensor and accelerometer.

Tilt Compensation: Module that calculates heading angle (yaw) by receiving the output of the open source Madgwick and Mahony filter algorithms (roll, pitch) and magnetometer sensing data.

4.2 Algorithm

4.2.1 Overview

The detection algorithm consists of three parts,

- Pill Event Detection,
- Swallow Confirmation Detection and
- Event Logging.

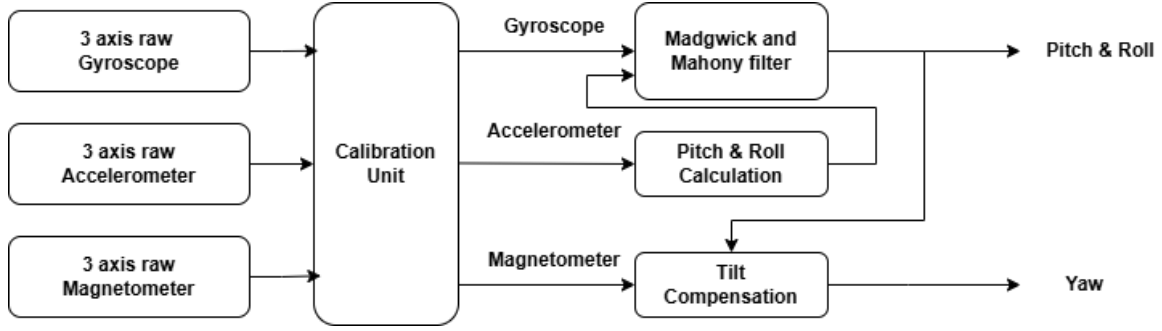


Figure 5 MPU9250 Sensor Fusion

where the Pill Event Detection is responsible for the pill retrieval detection and the swallow confirmation requires the patient actively confirm that he/she has swallowed the pill after retrieving the pill. After the two above detections finished, the detection results are logged for further statistics analysis. Figure 6 describes the overview of the algorithm.

4.2.2 Pill Event Detection

The pill event detection relies on four detectors:

- Pick-place detector,
- tilt detector,
- grasp detector and
- bottle opening detector.

Pick-place detector The pick-place detector requires sensing data from the accelerometer from IMU. Figure 7 shows principle of pick-place detector. The data from the accelerometer is monitored. The acceleration is computed as

$$a = \sqrt{ax^2 + ay^2 + az^2}, \quad (1)$$

where the ax , ay , and az are the acceleration along the x,y, and z axis measured by the IMU, respectively.

Data smoothing is an important part of the pick-place detector. If the data is not smoothed, the pick-place event will be very sensitive such that even small measure disturbances can trigger the event. The smoother takes the median of the last twenty(or less) data points as a baseline and computes the deviation between the new incoming data and the baseline. When an average of five deviations exceeds the pre-defined threshold, a motion activity is detected. The motion activity works as a flag and

only is true when the motion is detected. Otherwise, it remains false. The state change of the motion activity indicates the pick or place motion. If the state changes from stationary state to motion state, it means the container is being picked up. After the container is placed, a state change from the motion state to the stationary state can be easily detected and marked as a place state.

The pick state and the place state form a time window in which the sub-detectors play a crucial role.

Tilt detector The tilt detector monitors if the container is tilted during the pick-place time window for retrieving the pill. The tilt detector first initializes a rest gesture when the algorithm is started, and it assumes the container is in a stationary state at the beginning of the algorithm. The threshold builds a cone in 3D space. Figure 8 illustrates the cone of the threshold. The pose inside the cone is considered a rest pose, and the tilt pose outside.

Grasp & long press detector The grasp& long press detector relies on the pressure sensor on the side of the container. It is sensitive to the small pressure change on the bottle side. Two thresholds are pre-defined to detect the grasp and long press. If the pressure is higher than a relatively low threshold during the pick-place time window, a grasp activity is detected. In another condition, when the pressure sensor is pressed with a relatively higher force and held for at least one second during the swallow event, an active confirmation is logged to ensure the patient has swallowed the pill.

Bottle opening detector The light intensity sensor is responsible for the bottle opening detector. The sensor is put inside the bottle, and once the bottle is opened during the pick-place time window, a bottle-opening activity is marked.

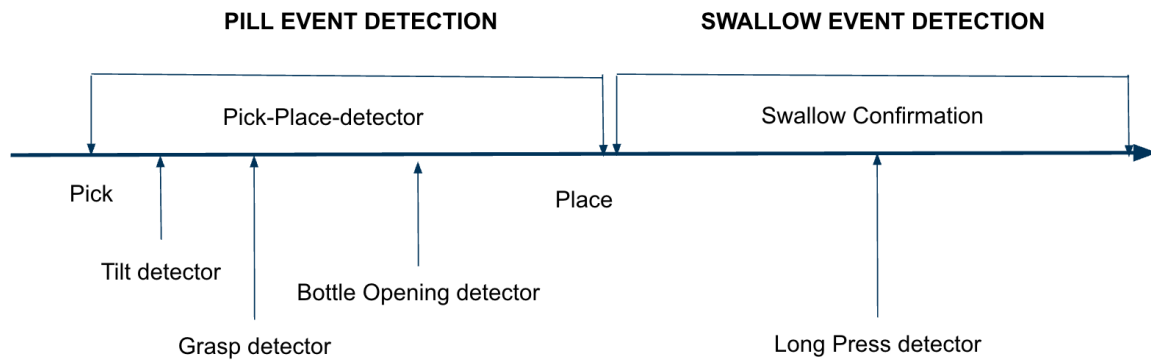


Figure 6 An overview of the detector algorithm: The algorithm contains pill event detection and swallow event detection consisting of multiple detectors.



Figure 7 Principle of Pick-place detector: The pick-place detector detects mainly the acceleration change using an accelerometer; a great acceleration change indicates a motion state; otherwise, a stationary state is detected.

Pill event detection The pill event is monitored after the algorithm starts. The event detection is triggered automatically by the detection of pick-place activity. All sensory data is saved in a buffer. When the pick-place time window is confirmed, the sensory data read within the time window is processed. If and only if the tilt, grasp, and bottle-opening activities are detected, a pill event is finally detected. In addition, a swallow event detection is automatically started after the pill event is detected.

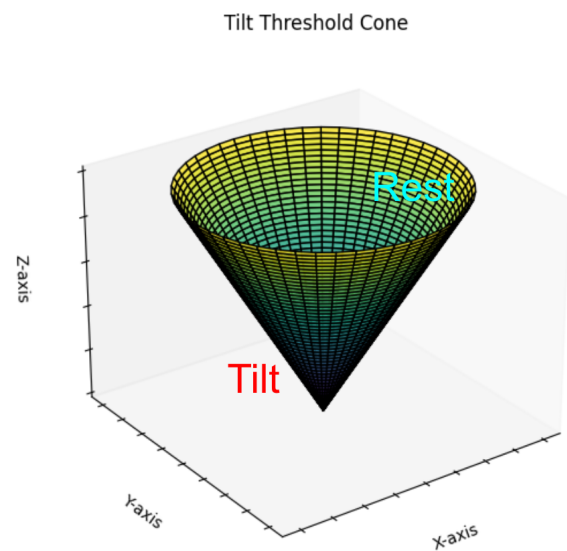


Figure 8 Tilt Threshold Cone: the threshold builds a cone in 3D space. It indicates rest pose within the cone and tilt outside the cone.

4.2.3 Swallow event detection

The swallow event is monitored after the pill event. It waits for ten seconds for the confirmation of the patient. If the patient does not actively confirm, a possible forget is logged, indicating the patient possibly forgot to swallow the pill after the retrieval of the pill.

4.2.4 Event logging

Every event detection is logged comprehensively for the recording and monitoring of medical adherence. Listing 1 logs a successful cycle of detection.

Listing 1 Log of a successful attempt

```
2025-03-01 21:59:17 - PILL EVENT DETECTED
Pill Event: Time 0m 21.4s
- Within Pick-Place window: 20.4s to 24.6s
- 3D Tilt: 28.18
- Orientation: Roll=-18.58, Pitch=0.02, Yaw=66.21
- Pressure: 552 at 24.4s
- Light: 182.50 lux at 21.4s
- Awaiting swallow confirmation (press and hold pressure sensor within 10s)
2025-03-01 21:59:17 - Actual time of event: 2025-03-01 21:59:17
2025-03-01 21:59:18 - SWALLOW CONFIRMED
PILL SWALLOW CONFIRMED: Time 0m 26.1s
- Pill taken at: 21.4s
- Confirmation delay: 4.6s
2025-03-01 21:59:18 - Actual time of swallow confirmation: 2025-03-01 21:59:18
```


Besides, a dynamic plot intuitively shows the events as figure 9. The plot shows all sensory input with the event areas. The yellow shadowed area indicates a fraud attempt, which triggered a tilt detector, but the pick-place precondition is not fulfilled. The green shadowed area describes a successful and complete attempt with pill event and swallow confirmation. The red shadowed area means a failed attempt with pill event yet without a swallow confirmation.

4.3 ROS Package

To show the potential of connecting the smart pill container to any other network devices and demonstrate its ability to be part of IoT, a ROS package is developed. Robot Operating System(ROS) provides a platform that controls the robot. The robot and host computer communicate with each other through ROS and the network. Since the robot-environment interaction becomes important, putting the smart items into the ROS framework shows its huge potential of being a part of the home automation systems in the future.

ROS Message All items in the ROS framework communicates through ROS message, which is a specific form of message. It is crucial to pre-define the format of the ROS message for the devices to integrate into ROS. A pre-defined example of ROS message is listed below as listing 2.

Listing 2 ROS message of PillEvent.msg

```
# PillEvent.msg - For pill retrieval events
Header header
time event_time
float32 tilt
float32 roll
float32 pitch
float32 yaw
float32 pressure
float32 light
time pressure_time
time light_time
time pick_time
time place_time
bool confirmed
bool forgotten
```

ROS Topic If the ROS Message is the format of the information in ROS, the ROS Topic is the medium of the ROS Message. The devices publish

their information through ROS Topic as the format of ROS Message. Figure 10 illustrates the information being published to the kernel of the ROS for further processing by other devices.

4.4 Statistical Analysis

To better demonstrate its functionality, we conducted a statistical simulation using synthetic medication adherence data for January, February, and March. By setting predefined dosage schedules (e.g., three times a day or every eight hours), we generated detailed charts to offer valuable insights into patients' medication habits.

These charts record medication intake throughout the day, categorizing the status into confirmed intake, forgotten doses, or missed doses. They also track daily adherence over a month. Three different statistical visualization methods are used to intuitively display monthly medication adherence records.

The January data, as shown in the figure 11, is presented using scatter plots and bar charts, with detailed statistics recorded in a table. The overall adherence rate was 91.4%, indicating good compliance. Out of 31 days, 24 days had perfect adherence, while 7 days had at least one missed pill. The most common issue was missing a single pill rather than skipping medication entirely. A time-of-day analysis revealed occasional missed doses, with some instances of potential forgotten pills, suggesting areas where adherence could be improved.

The February data, as shown in the figure 12, illustrated through a medication adherence calendar, provides a clear visualization of daily adherence rates over a five-week period using a color-gradient heatmap. Most days showed adherence above 80% (green), indicating strong consistency. However, specific dates—February 3, 7, 17, and 27—stood out with adherence rates below 40% (orange). These fluctuations suggest irregularities on certain days, potentially caused by routine disruptions or other external factors, warranting further investigation.

The March data, as shown in the figure13, provides a detailed breakdown of medication adherence by categorizing doses into three time periods—morning, afternoon, and evening—as well as by hourly distribution. The analysis revealed a clear adherence pattern throughout the day.

Morning adherence was the highest, with 93.5% of doses confirmed, suggesting that patients were

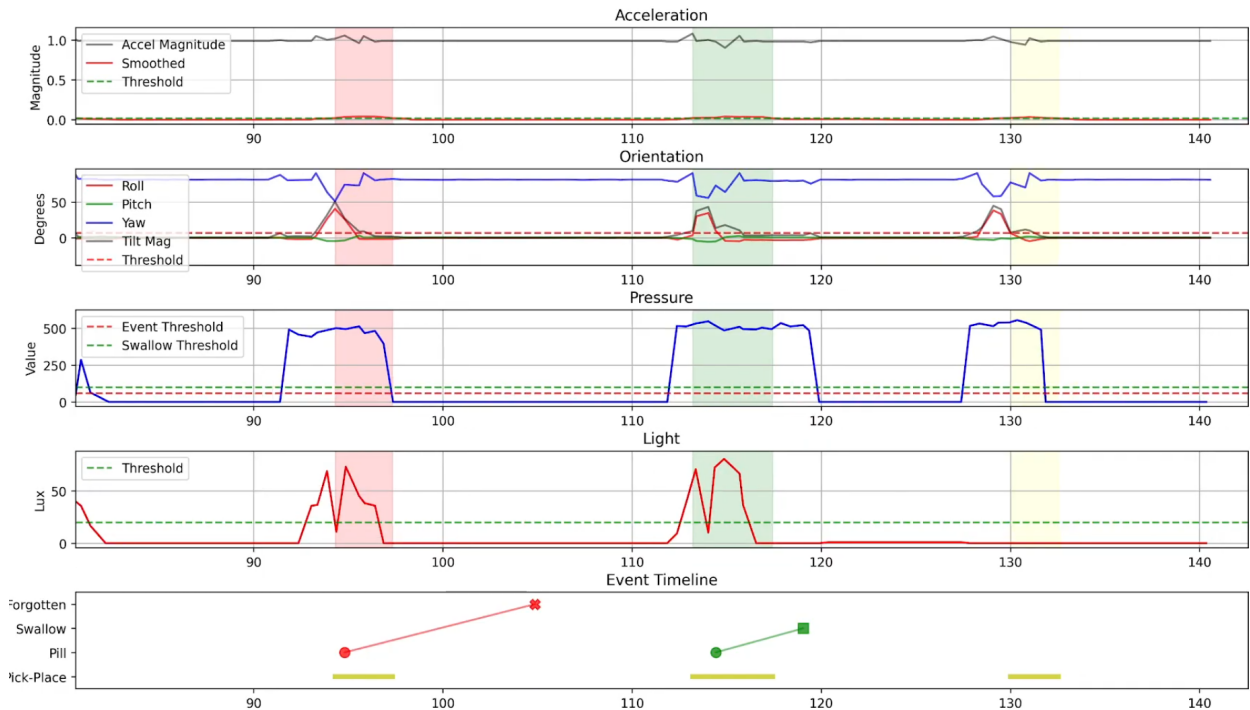


Figure 9 Dynamic plot with a failed, a fraud and a successful attempt: The first four rows illustrate all sensor data and threshold. The last row records the timeline of the event. The red shadowed area means the pill event is marked, but the patient has not confirmed the swallow, which indicates a failed attempt. The yellow shadowed area means the tilt activity but the whole pill event is undetected, which expresses a fraud attempt. The green shadowed area means a successful attempt with pill event and swallow confirmation, which is a successful attempt.

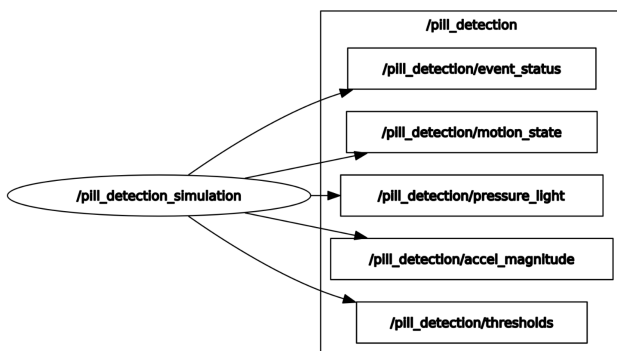


Figure 10 ROS Topics of the package: The smart pill container publishes all useful information through ROS Topic for other devices.

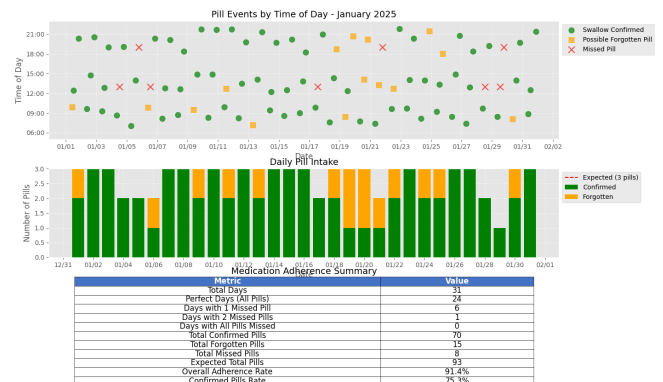


Figure 11 Statistical Simulation of Medication Adherence – January

most consistent in taking their medication early in the day. However, adherence dropped significantly in the afternoon to 77.4%, with a noticeable increase in both forgotten and missed doses, indicating a potential risk period. Evening adherence improved slightly to 80.6% but remained lower than morning intake.

A more granular hourly analysis pinpointed 13:00 and 19:00 as the most problematic times, where

the highest number of missed or forgotten pills occurred.

This helps identify medication patterns or anomalies. These visualizations provide a comprehensive overview of adherence trends, enabling better assessment of patient compliance and potential areas for intervention.

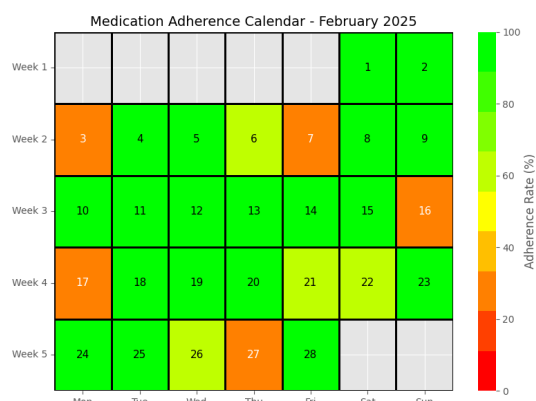


Figure 12 Statistical Simulation of Medication Adherence – February

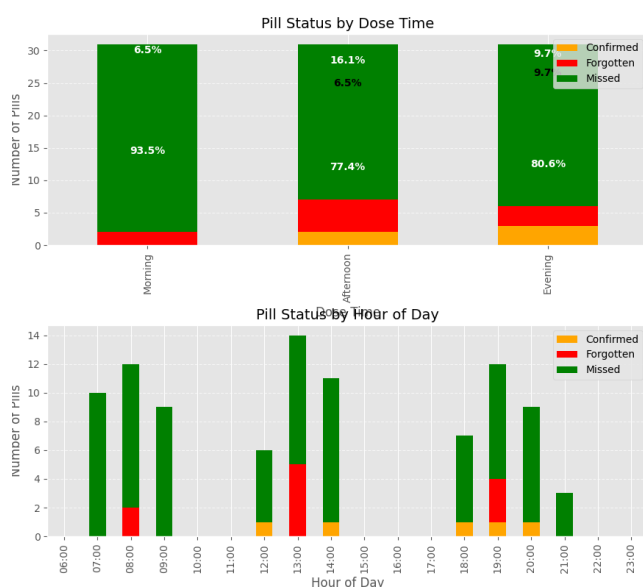


Figure 13 Statistical Simulation of Medication Adherence – March

5 Conclusion

Our smart pill container integrates a variety of sensors and algorithms to ensure effective detection of medication-taking behaviors, and it presents medication adherence through multiple data formats such as charts, calendars, and statistical reports. These data not only provide patients with clear feedback on their medication habits but also offer valuable analytical tools for healthcare teams. With this system, we hope to improve medication adherence and provide valuable insights for healthcare professionals.

Furthermore, our cost-effective design ensures accessibility to a wide range of users, making ad-

vanced medication adherence solutions more feasible on a large scale.

Looking ahead, we are committed to refining our system through sensor upgrades, enhanced data analytics, and broader IoT interoperability. By continuously improving its functionality, we aim to make our smart pill container a staple in medication management, ultimately contributing to better healthcare outcomes worldwide.

6 Future Work

One of our key next steps is to enhance the smart pill container's connectivity by integrating it with a Raspberry Pi or similar microcontroller. This upgrade will enable the wireless transmission of adherence data to cloud-based healthcare platforms, allowing for real-time monitoring, automated alerts, and timely interventions. Therefore, we are developing a ROS Noetic (Robot Operating System) package to demonstrate the Smart Pill Container's potential for seamless integration with other networked devices. This advancement highlights its future role in home automation systems, where sensor measurements and event statuses can be published as ROS topics, enabling effortless communication with smart home ecosystems and healthcare monitoring platforms.

To further improve accuracy, we plan to incorporate a weight sensor to precisely detect changes in the pill count. This addition will help ensure that the correct dosage is taken, distinguishing between proper medication use and accidental overuse.

Additionally, we aim to leverage machine learning models to analyze patient behavior over time. Through AI-driven predictive analytics, the system will identify potential non-adherence risks, allowing for proactive intervention and personalized reminders.

Finally, we will develop a user-friendly mobile application to provide patients and caregivers with real-time access to adherence records, medication reminders, and instant feedback on medication intake, further enhancing the system's usability and effectiveness.

References

- [1] Caroline A Walsh, Caitriona Cahir, Sarah Tecklenborg, Catherine Byrne, Michael A Culbertson, and Kathleen E Bennett. The association between medication non-adherence and adverse health outcomes in ageing populations: a systematic review and meta-analysis. *British journal of clinical pharmacology*, 85(11):2464–2478, 2019.
- [2] Marie T Brown and Jennifer K Bussell. Medication adherence: Who cares? In *Mayo clinic proceedings*, volume 86, pages 304–314. Elsevier, 2011.
- [3] Rachelle Louise Cutler, Fernando Fernandez-Llimos, Michael Frommer, Charlie Benrimoj, and Victoria Garcia-Cardenas. Economic impact of medication non-adherence by disease groups: a systematic review. *BMJ open*, 8(1):e016982, 2018.
- [4] S Thangam, T Sudeep Reddy, T Krithin, J Jesy Janet Kumari, et al. A smart pill container for improved medication. In *2024 5th International Conference on Electronics and Sustainable Communication Systems (ICESC)*, pages 1623–1630. IEEE, 2024.
- [5] Murtadha Aldeer, Richard E Howard, Richard P Martin, and Jorge Ortiz. Unobtrusive patient identification using smart pill-bottle systems. *Internet of Things*, 14:100389, 2021.
- [6] Tanja R Zijp, Daan J Touw, and Job FM van Boven. User acceptability and technical robustness evaluation of a novel smart pill bottle prototype designed to support medication adherence. *Patient preference and adherence*, pages 625–634, 2020.
- [7] Pradyumna Gokhale, Omkar Bhat, and Sagar Bhat. Introduction to iot. *International Advanced Research Journal in Science, Engineering and Technology*, 5(1):41–44, 2018.
- [8] Fahmi Sharmila Mat Nasir, Omar Abu Hassan, and Muhammad Rusydi Muhammad Razif. Smart medication adherence system using iot for elderly patients. *Progress in Engineering Application and Technology*, 3(1):559–570, 2022.
- [9] Rania A Molla. A smart prescription container for monitoring chronic disease patients. In *2023 IEEE International Smart Cities Conference (ISC2)*, pages 01–06. IEEE, 2023.
- [10] Dana DeMeo and Michael Morena. Medication adherence using a smart pill bottle. In *2014 11th International Conference & Expo on Emerging Technologies for a Smarter World (CEWIT)*, pages 1–4. IEEE, 2014.
- [11] Roisul Islam Rumi, Monirul Islam Pavel, Ekhwan Islam, Mohsinul Bari Shakir, and Mohammad Amzad Hossain. Iot enabled prescription reading smart medicine dispenser implementing maximally stable extremal regions and ocr. In *2019 Third International conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud)(I-SMAC)*, pages 134–138. IEEE, 2019.
- [12] Nurmiza Binti Othman and Ong Pek Ek. Pill dispenser with alarm via smart phone notification. In *2016 IEEE 5th global conference on consumer electronics*, pages 1–2. IEEE, 2016.
- [13] Vinay Peddisetti, Pavan Kumar Kandregula, Joel Anil John, Suseela Poomdla, Kiran George, and Anand Panangadan. Smart medication management: Enhancing medication adherence with an iot-based pill dispenser and smart cup. In *2024 IEEE First International Conference on Artificial Intelligence for Medicine, Health and Care (AIMHC)*, pages 137–144. IEEE, 2024.
- [14] Yosef Houmid Al Shammari, Homoud Awade Al Shammari, Talal Muhammad Bin Barak, Bandar Said Alharbi, Saud Tawfiq Alshammari, Saleh Abdullah Alnais, Saeed Awad Aljohani, Abdulrahman Abdullah Alorf, Meshari Abdulmajeed Alnawmasi, Mohammed Awad Alshammari, et al. Smart pills: the future of adherence monitoring. *International journal of health sciences*, 3(S1):1–15.
- [15] kriswiner. Mpu9250. <https://github.com/kriswiner/MPU9250>, 2025.